

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **ML models** for user personalization, achieving a **15% uplift** in click-through rates by developing a collaborative filtering recommendation engine.
- Engineered scalable **ML pipelines** using **Python** and **Spark** for data processing, model training, and deployment, reducing model retraining time by **40%**.
- Conducted rigorous **A/B tests** to evaluate ML model performance, optimizing personalization algorithms and contributing to a **10% increase** in user engagement.
- Analyzed large datasets with **SQL** and **pandas** to extract actionable insights on user acquisition and retention, informing strategic decisions that improved conversion rates by **8%**.
- Collaborated with cross-functional teams to translate business requirements into technical solutions, integrating **ML models** into growth initiatives and enhancing user experience.
- Contributed to the development of the **ML infrastructure**, ensuring support for rapid experimentation and deployment of new models, enhancing deployment frequency by **25%**.

SKILLS

Machine Learning – Recommendation Systems, Personalization Algorithms, Model Training, A/B Testing, Statistical Analysis, PyTorch, scikit-learn.

Data Engineering – SQL, Spark, Hadoop, pandas, NumPy, Data Pipelines, ETL.

Programming – Python, Java, Spring Boot, REST APIs.

MLOps – Model Deployment, CI/CD, Docker, Kubernetes.

Cloud & Infrastructure – AWS, EC2, S3, Lambda, CloudWatch.

Development Tools – Git, Jenkins, Jira, Confluence.

Data Visualization – Power BI, Tableau, R.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Personalized Content Recommendation Engine (Pinterest Growth)

- Developed a **real-time recommendation engine** using **PyTorch** and **Python** to surface personalized content suggestions to millions of users, increasing engagement by **12%**.
- Built scalable **ML pipelines** with **Spark** for feature engineering and model training on terabytes of user interaction data, reducing training cycles from days to hours.

- Implemented **A/B testing** frameworks to compare model performance against baseline strategies, resulting in a **15%** improvement in key user growth metrics and validating algorithmic enhancements.
- Integrated the recommendation service via **REST APIs** with the main platform, ensuring low latency and high availability for a seamless user experience.
- Managed the deployment of the ML model into a **Docker** container orchestrated by **Kubernetes** on **AWS**, enabling efficient scaling and automated updates.

Financial Services Transaction Monitoring Dashboard (HCL Technologies)

- Engineered a **payments operations dashboard** using **Java Spring Boot** and **PostgreSQL** to monitor transaction flows and identify anomalies in near real-time.
- Automated alert generation for failed or stuck payment messages, reducing manual intervention by **30%** and improving incident response times.
- Collaborated with business analysts to define requirements for transaction monitoring and reconciliation workflows, ensuring adherence to regulatory awareness standards.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)